

Girraween High School
Mathematics

Year 11

June 2004

Task 3

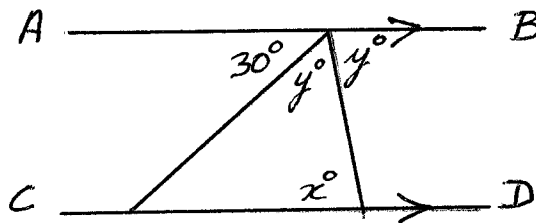
Time: 90 minutes

Instructions:

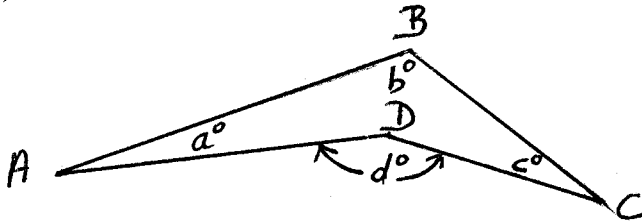
1. Attempt all questions.
2. Write your answers on your own paper.
3. All necessary working must be shown.
4. Marks will be deducted for careless or badly arranged work.

QUESTION 1 (21 marks)

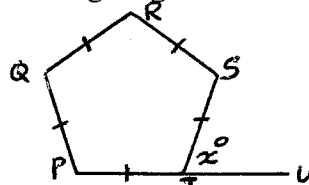
- (a) Find the value of x and y . (Give reasons).



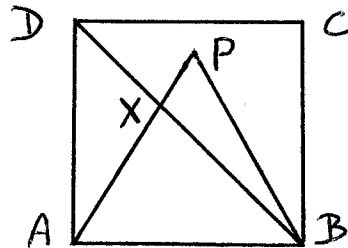
- (b) Prove that $d = a + b + c$.



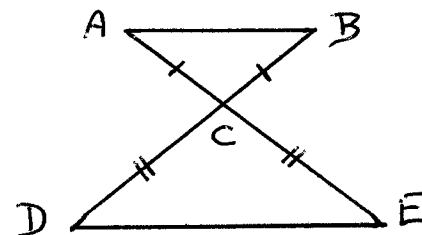
- (c) Find the value of x giving reasons.



- (d) In the diagram ABCD is a square and PAB is an equilateral triangle. Calculate the size of the angle DXA in degrees, giving reasons.

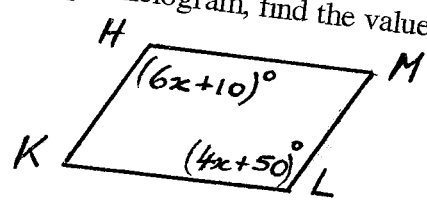


- (e) Given that $AC = CB$ and $DC = CE$, prove that AB is parallel to DE.

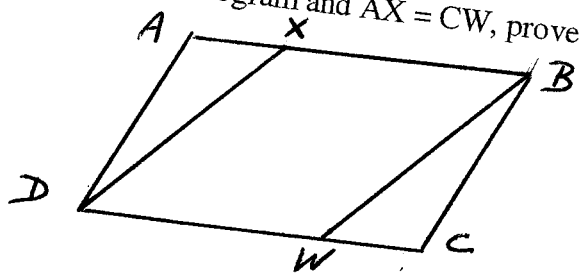


QUESTION 2 (21 marks)

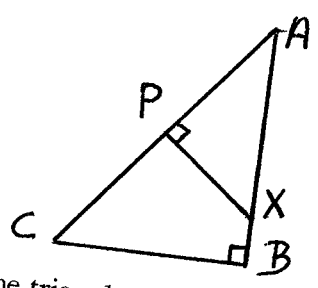
- (a) If KHML is a parallelogram, find the value of x , giving reasons.



- (b) If ABCD is a parallelogram and $AX = CW$, prove that $XD = BW$.



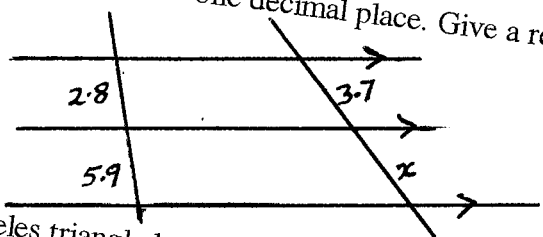
(c)



- (i) Show that the triangles PAX and BAC are similar.
 (ii) If AX and BC are both 5 cm, and AP is 4 cm, find PX and AC.
 (d) The diagonals of a rhombus measure 8 cm and 15 cm. Calculate :
 (i) the length of each side.
 (ii) the area of the rhombus.

QUESTION 3 (15 marks).

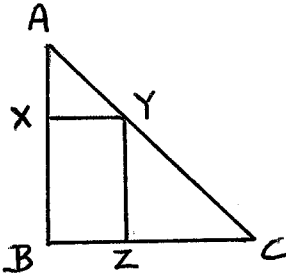
- (a) Find the value of x to one decimal place. Give a reason for your answer.



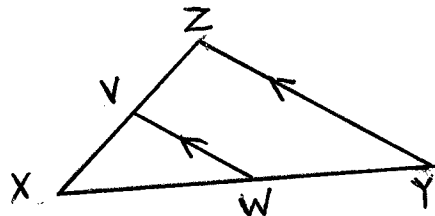
- b) An isosceles triangle has two sides 12 cm long and a perpendicular height of 10 cm.
 Find the length of the base. (Leave answer in exact form).

Question 3 (continued)

- (c) In the diagram XYZB is a rectangle. If $AB = 8$ cm, $YZ = 5$ cm, $XY = 3$ cm, find the length of BC.



- (d) In the figure, $XV = 12$ cm, $XY = 20$ cm, $VZ = 18$ cm and WV is parallel to YZ . Find the length of XW . (Give reasons for your answer).



QUESTION 4 (20 marks).

- (a) If $f(x) = x^2 + 3x - 4$ find the values of :

(i) $f(0)$ (ii) $f(-2.5)$ (iii) $f(x - 1)$

- (iv) If $u(x) = f(x) - f(x - 1)$, show that

(α) $u(x) = 2x + 2$ and (β) $f(2) = u(2)$.

- (b) A function is defined by the following rule :

$$f(x) \begin{cases} = 0 & \text{if } x \leq -3 \\ = -1 & \text{if } -3 < x < 0 \\ = x & \text{if } x \geq 0 \end{cases}$$

Find : (i) $f(-3) + f(-2) + f(2)$ (ii) $f(a^2)$.

- (c) Determine whether these functions are even, odd or neither. (Give reasons).

(i) $y = 1 + 2x^2$ (ii) $y = (1 + x^5)^2$

- (d) Find the equation of a circle with centre $(-3, 2)$ and radius 2 units.

QUESTION 5 (17 marks).

(a) Draw a neat sketch for the following :

(i) $y = |x| + 1$ (ii) $y = -\sqrt{9 - x^2}$ (iii) $y = 2^x$.

(b) State whether or not the following are functions :

(i) $x^2 + y^2 = 9$ (ii) $y = -x^2$ (iii) $y = \frac{1}{x}$.

(c) State the domain and range for the following :

(i) $y = \frac{1}{x}$ (ii) $y = 2^x$ (iii) $y = x^2 + 2$

(d) State the domain of :

(i) $y = \sqrt{x+1}$ (ii) $y = \sqrt{9 - x^2}$

END OF TASK.

Year 11 Mathematics Task 3, June 2004. Solutions.

1. (a) $2y + 30 = 180$ (straight $\angle$)

$$\therefore 2y = 150$$

$$\therefore y = 75 \# \quad (3)$$

$$x = y \text{ (alternate } \angle\text{s, } AB \parallel CD)$$

$$\therefore x = 75 \# \quad (3)$$

(b) $\angle ADC = 360 - d$ ($\angle$ s at a point)

$$\therefore a + b + c + (360 - d) = 360 \text{ (sum of a quad.)}$$

$$\therefore a + b + c = d \# \quad (4)$$

(c) $5x = 360$ (sum of exterior $\angle$ s)

$$\therefore x = 72 \# \quad (3)$$

OR

$\angle$ sum of pentagon

$$= (5-2) \times 180^\circ = 540^\circ$$

$$\therefore \text{each interior } \angle = \frac{540^\circ}{5} = 108^\circ$$

$$\therefore x + 108 = 180 \text{ (straight } \angle)$$

$$\therefore x = 72 \#$$

(d) $\angle PAB = 60^\circ$ ($\angle$ s of equilateral $\triangle APB$)

$$\angle PAD = 30^\circ \text{ (}\angle DAB \text{ is } 90^\circ \text{ in a square)}$$

$$\angle ADB = 45^\circ \text{ (diagonal BD bisects } \angle ADC)$$

$$\therefore \angle DXA + \angle PAD + \angle ADB = 180^\circ \text{ (}\angle \text{ sum of } \triangle)$$

$$\therefore \angle DXA + 30^\circ + 45^\circ = 180^\circ$$

$$\angle DXA + 75^\circ = 180^\circ$$

$$\therefore \angle DXA = 105^\circ \# \quad (4)$$

(e) In $\triangle$ s ACB and ECD

$$\angle ACB = \angle ECD \text{ (vertically opposite } \angle\text{s)}$$

$$\frac{AC}{CE} = \frac{BC}{DC} \text{ (same ratio)}$$

$$\therefore \triangle ACB \parallel \triangle ECD \text{ (two sides of one } \triangle \text{ are proportional to 2 sides of another } \triangle \text{ and included } \angle\text{s are equal)}$$

$$\therefore \angle ABC = \angle EDC \text{ (corresponding } \angle\text{s of similar } \triangle\text{s)}$$

$$\text{But there are alternate angles, } \therefore AB \parallel DE \# \quad (4)$$

2. (a) $6x + 10 = 4x + 50$ (opposite $\angle$ s of $\parallel$ gram)

$$\therefore 6x - 4x = 50 - 10$$

$$2x = 40$$

$$\therefore x = 20 \# \quad (3)$$

(b) In $\triangle$ s AXD and CWB

$$AX = CW \text{ (given)}$$

$$AD = CB \text{ (opposite sides of } \parallel \text{ gram)}$$

$$\angle XAD = \angle WCB \text{ (opposite } \angle\text{s of } \parallel \text{ gram)}$$

$$\therefore \triangle AXD \equiv \triangle CWB \text{ (SAS)}$$

$$\therefore XD = WB \text{ (corresponding sides of congruent } \triangle\text{s)}$$

$$(5) \#$$

2 (c) (i) In Δ s PAX and BAC

$\angle A$ is common
 $\angle P = \angle B$ (right $\angle$)
 $\therefore \Delta PAX \parallel \Delta BAC$ (equiangular)

(ii) $PX^2 = 5^2 - 4^2$
 $= 25 - 16$
 $= 9$

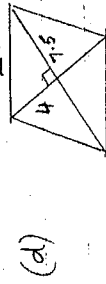
$\therefore PX = 3 \text{ cm}$ #

$\frac{AX}{AC} = \frac{PX}{BC}$ (same ratio)

$\therefore \frac{5}{AC} = \frac{3}{5}$ (3)

$\therefore 3AC = 25$

$\therefore AC = \frac{25}{3}$ or $8\frac{1}{3} \text{ cm}$ #



(i) $x^2 = 4^2 + 7.5^2$
 $= 16 + 56.25$
 $= 72.25$

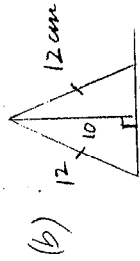
$\therefore x = \sqrt{72.25} = 8.5 \text{ cm}$ #

(ii) Area = $\frac{1}{2} \times 8 \times 15 = 60 \text{ cm}^2$ #

3. (a) $\frac{2.8}{5.9} = \frac{3.7}{x}$ (ratios of intercepts)

$2.8x = 3.7 \times 5.9$
 $x = \frac{3.7 \times 5.9}{2.8}$

$x = 7.8$ (1 dec. pl.) #



$y^2 = 12^2 - 10^2 = 144 - 100$
 $y = \sqrt{44} = 2\sqrt{11}$

$\therefore \text{base} = 2y$
 $= 4\sqrt{11} \text{ cm}$ #

(c) In Δ s AX'Y and ABC
 $\angle A$ is common

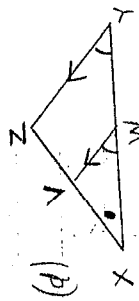
$\angle AXY = \angle ABC = 90^\circ$

$\therefore \Delta AXY \parallel \Delta ABC$ (equiangular)

$\therefore \frac{AX}{AB} = \frac{XY}{BC}$ (corresponding sides in same ratio)

$\therefore \frac{3}{8} = \frac{3}{BC}$

$\therefore BC = 8 \text{ cm}$ #



In Δ s XWV and XYZ
 $\angle X$ is common

$\angle W = \angle Y$ (corresponding $\angle$ s, $VW \parallel ZY$)

$\therefore \Delta XWV \parallel \Delta XYZ$

$\frac{XW}{XY} = \frac{XV}{XZ}$ (corresponding sides in same ratio)

$\therefore \frac{XW}{20} = \frac{12}{30}$

$\therefore 5XW = 40, \therefore XW = 8 \text{ cm}$ #

4. (a) $f(x) = x^2 + 3x - 4$

(i) $f(0) = -4$ # (1)

(ii) $f(-\frac{5}{2}) = (-\frac{5}{2})^2 + 3(-\frac{5}{2}) - 4$
 $= \frac{25}{4} - \frac{15}{2} - 4$
 $= -5\frac{1}{4}$ # (2)

(iii) $f(x-1) = (x-1)^2 + 3(x-1) - 4$
 $= x^2 - 2x + 1 + 3x - 3 - 4$
 $= x^2 + x - 6$ # (2)

(iv) $u(x) = f(x) - f(x-1)$

(a) $u(x) = x^2 + 3x - 4 - (x^2 + x - 6)$
 $= x^2 + 3x - 4 - x^2 - x + 6$
 $= 2x + 2$ # (2)

(b) $f(2) = 2^2 + 3(2) - 4 = 6$
 $u(2) = 2(2) + 2 = 6$
 $\therefore f(2) = u(2)$ # (2)

(b) (i) $f(-3) + f(-2) + f(2)$
 $= 0 + (-1) + 2$
 $= 1$ # (2)

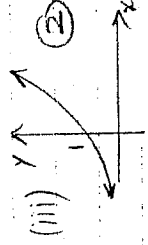
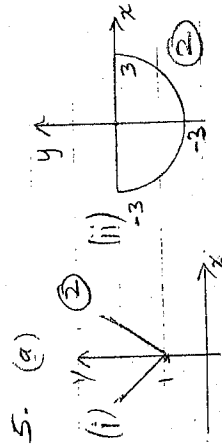
(ii) $f(a^2) = a^2$ # (1)

(c) (i) $f(x) = 1 + 2x^2$
 $f(-x) = 1 + 2(-x)^2$
 $= 1 + 2x^2$

$f(x) = f(-x) \therefore$ even function. #

(ii) $f(x) = (1+x)^2$
 $f(-x) = (1-x)^2$
 $-f(x) = -(1+x)^2$
 $\therefore$ neither odd nor even function. # (3)

(d) $(x+3)^2 + (y-2)^2 = 4$ # (2)



(b) (i) NO (ii) YES (iii) YES (3)

(c) (i) D: $\{x: \text{all } x, x \neq 0\}$
R: $\{y: \text{all } y, y \neq 0\}$ (1)

(ii) D: all real x
R: $y > 0$ (1)

(iii) D: $\{x: \text{all real } x\}$
R: $\{y: y > 2\}$ (1)

(d) (i) D: $\{x: x \geq -1\}$ (1)

(ii) D: $\{x: -3 \leq x \leq 3\}$ (1)